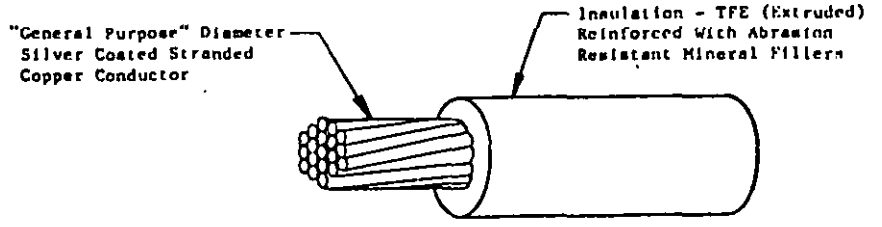
MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, ABRASION RESISTANT, EXTRUDED TFE, MEDIUM WEIGHT, SILVER-COATED COPPER CONDUCTOR, 600-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759.




TFE - Polytetrafluoroethylene

Note: Outer Surface Must be Smooth, Homogeneous, Virgin TFE With no Mineral Filler Exposed

TABLE I. CONSTRUCTION DETAILS

| Part No. 1/ | Wire size | Stranding (Number of strands X AWG gage of strands) | Diameter of stranded conductor (inches) | | Finished wire | | |
|---|---|---|---|---|---|---|---|
| | | | (min) | (max) | Resistance at 20°C (68°F) (ohms/1000 ft) (max) | Diameter (inches) | Weight (lbs/1000 ft) (max) |
| M22759/7-24-* | 24 | 19 X 36 | .023 | .025 | 24.3 | .062 ±.002 | 4.3 |
| M22759/7-22-* | 22 | 19 X 34 | .029 | .032 | 15.1 | .073 ±.002 | 6.0 |
| M22759/7-20-* | 20 | 19 X 32 | .037 | .040 | 9.19 | .082 ±.002 | 8.1 |
| M22759/7-18-* | 18 | 19 X 30 | .046 | .050 | 5.79 | .092 ±.002 | 11.0 |
| M22759/7-16-* | 16 | 19 X 29 | .052 | .057 | 4.52 | .102 ±.003 | 13.8 |
| M22759/7-14-* | 14 | 19 X 27 | .065 | .072 | 2.88 | .115 ±.003 | 18.6 |
| M22759/7-12-* | 12 | (B) 19 X 25 | (B) .082 | (B) .090 | (B) 1.81 | .134 ±.003 | (B) 28.5 |
| M22759/7-10-* | 10 | 37 X 26 | .106 | .112 | 1.19 | .158 ±.004 | 41.8 |
| M22759/7-8-* | 8 | 133 X 29 | .158 | .169 | .658 | .220 ±.005 | 73.1 |
| M22759/7-6-* | 6 | 133 X 27 | .198 | .213 | .418 | .270 ±.006 | (B) 111. |
| M22759/7-4-* | 4 | 133 X 25 | .250 | .268 | .264 | .328 ±.007 | (B) 169. |

1/ PART NO.: The asterisks in the part number column, Tables I through III, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M22759/7-20-9; white with orange stripe - M22759/7-20-93.

(B) denotes changes



FSC 6145

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TABLE II. PERFORMANCE DETAILS

| Part No. | Abrasion resistance | | | | | Bend testing | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Mandrel diameter (inches) (±3%) | | Test load (lbs) (±3%) | |
| | Procedure | Resistance (inches of tape) (min) (initial and after immersion) | Weight support bracket | Weight (lbs) | Tension load (lbs) | Life cycle (oven & bend tests) 1/ | Cold bend test | Life cycle (oven & bend tests) 1/ | Cold bend test |
| M22759/7-24-* | II | 48 | A | 3.0 | 1.0 | 4.5 | 3.0 | .75 | 2.0 |
| M22759/7-22-* | II | 54 | A | 3.0 | 1.0 | 4.5 | 3.0 | .75 | 2.0 |
| M22759/7-20* | II | 66 | A | 3.0 | 1.0 | 4.5 | 3.0 | .75 | 2.0 |
| M22759/7-18-* | II | 72 | A | 3.0 | 1.0 | 4.5 | 3.0 | 1.00 | 2.0 |
| M22759/7-16-* | II | 81 | A | 3.0 | 1.0 | 6.5 | 3.0 | 1.00 | 3.0 |
| M22759/7-14-* | II | 81 | A | 4.25 | 2.0 | 6.5 | 3.0 | 3.00 | 3.0 |
| M22759/7-12-* | II | 81 | B | 4.25 | 2.0 | 6.5 | 6.0 | 3.00 | 3.0 |
| M22759/7-10-* | II | 81 | B | 4.25 | 2.0 | 10.0 | 6.0 | 3.00 | 5.0 |
| M22759/7-8-* | I | 25 | B | 3.0 | 2.0 | 10.0 | 6.0 | 3.00 | 6.0 |
| M22759/7-6-* | I | 25 | C | 3.0 | 2.0 | 10.0 | 6.0 | 3.00 | 6.0 |
| M22759/7-4-* | I | 33 | C | 4.25 | 2.0 | 10.0 | 6.0 | 6.00 | 10.0 |

1/ Also for bend tests after immersion

ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 200°C (392°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
SPARK TEST OF PRIMARY INSULATION: Not required
IMPULSE DIELECTRIC TEST: 8.0 kilovolts (peak), 100% test
INSULATION RESISTANCE: 50,000 megohms for 1000 ft (min)
WRAP TEST: "Wrap back" test required; no cracking
Oven temperature, 313 ±2°C (595.4 ±3.6°F)
BLOCKING: 260 ±2°C (500 ±3.6°F)
SHRINKAGE 0.125 inch max at 313 ±2°C (595.4 ±3.6°F)
WICKING (PROCEDURE I): 1.0% (max) weight increase
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 2500 volts (rms), 60Hz
THERMAL SHOCK: Oven temperature, 200 ±2°C (392 ±3.6°F)
Max change in measurement Sizes 24 through 12: 0.060 inch
Sizes 10 through 8: 0.100 inch
Sizes 6 through 4: 0.125 inch
FLAMMABILITY: Post-flame dielectric test not required
LIFE CYCLE: Oven temperature 313 ±2°C (595.4 ±3.6°F)
Dielectric test, 2500 volts (rms), 60Hz
DIELECTRIC TEST AFTER IMMERSION: 2500 volts (rms), 60Hz
ABRASION RESISTANCE AFTER IMMERSION: Specimens after immersion shall meet the same abrasion requirements as initial specimens; this test to be a quality conformance test subject to the same inspection level and acceptable quality level as the test for initial abrasion resistance.
HUMIDITY RESISTANCE: 50,000 megohms for 1000 ft, min insulation resistance after humidity exposure
SURFACE RESISTANCE: 500 megohm-inches (min), initial and final readings
SMOKE: 313°C (595.4°F)
ACID RESISTANCE: Required
Dielectric test 2500 volts (rms), 60Hz
COLOR: In accordance with MIL-STD-104, Class 1; white preferred



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ADDITIONAL REQUIREMENTS (Continued)

COLOR STRIPING OR BANDING DURABILITY: 250 cycles (500 strokes) (min), 500 grams weight
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
WIRE LENGTH REQUIREMENTS: Schedule A
SUPERSESSION DATA: The wire of this specification sheet, by part number, replaces and supersedes the wire of MS18000(ASG) (canceled) in accordance with Table III.

TABLE III. SUPERSESSION BY PART NUMBER

| Part number MS18000(ASG) | Part number MIL-W-22759/7 |
|---|---|
| MS18000-24-* | M22759/7-24-* |
| MS18000-22-* | M22795/7-22-* |
| MS18000-20-* | M22759/7-20-* |
| MS18000-18-* | M22759/7-18-* |
| MS18000-16-* | M22759/7-16-* |
| MS18000-14-* | M22759/7-14-* |
| MS18000-12-* | M22759/7-12-* |
| MS18000-10-* | M22759/7-10-* |
| MS18000-8-* | M22759/7-8-* |
| MS18000-6-* | M22759/7-6-* |
| MS18000-4-* | M22759/7-4-* |

Custodians:
Navy - AS
Army - EL
Air Force - 80

Preparing activity:
Navy - AS
(Project No. 6145-0670-5)

Review activities:
Navy - EC, OS
Army - ME, MI, MU
Air Force - 80, 85
DSA - IS
NSA

User activities:
Navy - MC
Army - AT, AV
Air Force - 11

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